- Compute Match — Local Development Guide
  - Prerequisites
  - First run
  - Demo accounts
  - Development modes
  - Make targets
  - Local URLs and ports
  - Configuration files
  - Everyday loop
  - Common problems

Compute Match — Local Development Guide

Prerequisites

- Git, Bash, Make, Docker Desktop with Compose v2, and NVIDIA network/VPN access.
- Either **Devbox** 0.18.3 or newer (supplies all tools) or installed tools: Node 22 (frontend), Python 3.13 and `uv` (backend and worker).
- Docker-only mode (`make onboard` , `make start`) needs no Node or Python on your laptop.

First run

```
make onboard      # build containers, prepare databases and identity, create demo accounts and
        inventory
make verify       # should finish with "status": "ready"
```

The first build can take 10–20 minutes. Without Make, run `./local-setup/onboard.sh` . If it fails, read `local-setup/.onboarding/setup.log` , fix the problem and rerun.

Open `http://localhost:3000` , click **Login**, enter a demo account email, then its generated password.

Demo accounts

Passwords are generated into `local-setup/.onboarding/accounts.md` (human guide) and `credentials.json` (machine-readable, includes the Keycloak admin login). These files are Git-ignored.

Experience	Account
Customer	`developer@customer-a.example.org`
Provider (NCP)	`seed@singtel.example.org`
Platform business admin	`business-admin@platform.example.org`
Operations	`ops@platform.example.org`

Normal onboarding creates 17 accounts; Singtel and Firmus provider accounts test isolation between NCPs, and colleague accounts test shared inventory inside one NCP. `make onboard ONBOARD_ARGS="--with-scenarios"` adds 22-account regression personas, including `pending@scenarios.example.org` (pending activation), `terms@scenarios.example.org` (terms not accepted), `restricted@scenarios.example.org` (active, no business role), `infra-manager@customer-a.example.org` and `tech-admin@customer-b.example.org`.

Test sign-ups with addresses such as `you@example.com`; `example.com` and `nvidia.com` auto-activate, personal domains are blocked, and local email delivery is disabled. A new sign-up may need a role assigned in Keycloak (**customer** realm → user → **Role mapping**), then sign out and in again.

`make onboard ONBOARD_ARGS="--no-seed"` creates accounts without demo inventory. Sample CSVs for upload testing are in `local-setup/fixtures/`.

Development modes

Command	Runs on your laptop	Runs in Docker
`make start`	nothing	frontend, backend, worker
`make frontend`	frontend	backend, worker
`make backend`	backend	frontend, worker
`make worker`	worker	frontend, backend
`make dev` (fullstack)	all three	none

PostgreSQL, Keycloak, Temporal and Redis always stay in Docker. Install dependencies for a mode with `make deps MODE=backend` (or `frontend`, `worker`).

Make targets

Target	Purpose
`make help`	List commands
`make onboard`	Configure, start and seed (`ONBOARD_ARGS` for options)
`make doctor`	Check prerequisites and configuration only
`make deps`	Install locked dependencies for a `MODE`
`make build`	Build local images
`make start` / `make stop`	Start all in Docker / stop everything but keep data
`make start-infra`	Start only supporting containers
`make status`	Show mode, processes and URLs
`make logs SERVICE=backend`	Follow logs (frontend, backend or worker)
`make verify`	Check accounts, permissions, inventory, login routing and worker polling
`make test`, `make test-backend`, `make test-frontend`, …	Unit tests
`make quality`	Lint, type checks and tests
`make clean`	Remove containers, volumes and images
`make obs-start` / `obs-stop` / `obs-logs`	Optional SigNoz telemetry stack

Local URLs and ports

Service	Address
Application	http://localhost:3000
Backend API docs (Swagger)	http://localhost:8082/docs

Service	Address
Keycloak	http://localhost:8080 (admin at `/admin`, realm `customer`)
Temporal UI	http://localhost:7600
Temporal gRPC	localhost:7233
PostgreSQL	localhost:5432
Redis	localhost:6379
Superset (optional)	http://localhost:8088
SigNoz (optional)	http://localhost:3301 (OTLP gRPC 4317, HTTP 4318)

Configuration files

Location	Use
`local-setup/.env`	Generated service settings and secrets
`local-setup/.onboarding/credentials.json`	Account credentials
`local-setup/.onboarding/accounts.md`	Account guide
`local-setup/.onboarding/dev/`	Host-mode settings, process state and logs
`local-setup/.onboarding/setup.log`	Onboarding diagnostics

You do not need to copy `.env.sample` or create service dotenv files.

Everyday loop

`make stop` at the end of the day keeps your data; `make start` resumes. Use `make status`, then `make logs SERVICE=...` when something looks wrong.

Common problems

Problem	Fix
Image or Maven downloads fail	Connect to the NVIDIA VPN
Port already in use	Stop the other checkout or process using the port
Wrong Node version	Use Node 22 or set `LOCAL_DEV_NODE`
Login redirects to a Docker hostname	Restore localhost public URLs in the generated settings
Signed up but see no features	Assign a realm role in Keycloak and sign in again
SigNoz shows no data	Finish SigNoz first-user setup, run `make obs-start`, restart apps with tracing; Docker Engine 25+ for container metrics
Everything is broken	`make clean && make onboard && make verify`